

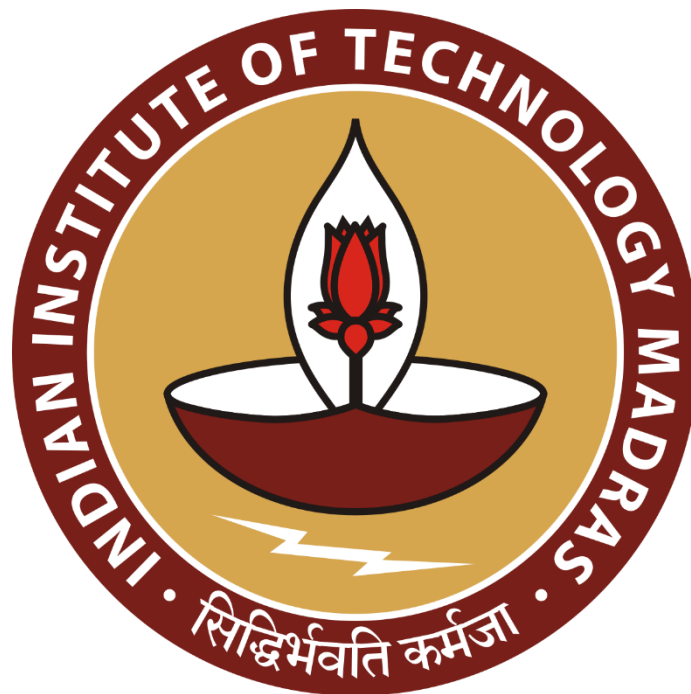
Enhancing Gym Profitability by Optimizing Member Retention & Acquisition

Mid-Term Report for the BDM capstone Project

Submitted by

Name: DAKSH

Roll number: 23F2001398



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1. Executive Summary and Title	3
2. Proof of originality of the Data	4
a. Letter from Organization with sign	
b. Images related to organization	
c. Recorded video interacting with the founder	
3. a. Metadata	5
b. Descriptive Statistics	
4. Detailed Explanation of Analysis Process / Method	8
5. Results and Findings	9

1. Executive Summary and Title

The Gym named F4 Fitness is located in Pochanpur Village, Sector-23 Dwarka, Delhi. It was established by Mr. Ajay Tomar in 2016. It operates in the field of health and physical fitness. The gym has employed the team of four personal trainers for these services.

The gym members are nearby residents and from all sections of society. This gym operates for twelve hours in two shifts (4 AM -10 AM and 4 PM –10 PM) for six days per week, excluding Sundays and gazetted holidays with an annual turnover of approximately 10 lakh rupees.

The major business issue that the organization is facing, is related to member retention & acquisition and it also has issue of overcrowding at specific time due to which members start leaving.

He also mentioned how the schools and college vacations, exams affect their business, how season affect their business like when there are vacations of schools and colleges, then the members increase and in the winter season, the members also increase and when there are exams, the number of members decreases.

The mid-term report contains proof of originality of data through relevant photos, videos and signed letter from organization, metadata section contains detailed explanations for each column along with link to the dataset, analysis process, results and findings.

Goal:

1. To find out why they are facing difficulties in membership acquisition and retention.
2. To find out the solution to the problem of overcrowding at certain period of times due to which members start leaving.
3. To find out what are the peak months of the business.

2. Proof of Originality of data

- **Organization's Name:** F4 Fitness Gym
- **Owner's Name:** Mr. Ajay Tomar
- **Owner's Phone Number:** +91 9910381877
- **Location:** Pochanpur Village, Sector-23 Dwarka, Delhi

Data is provided by Mr. Ajay Tomar who is the owner of F4 Fitness Gym.

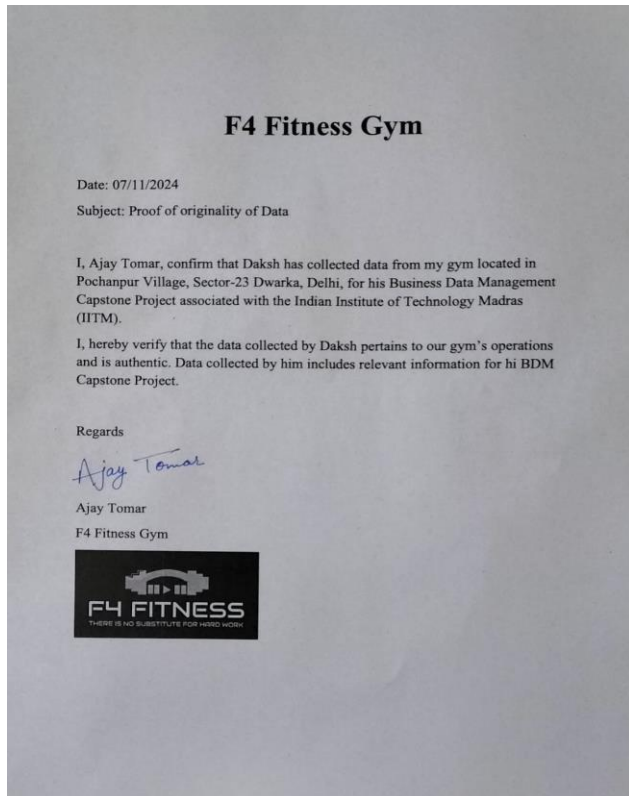


Fig 1. Letter from organization with sign



Fig 2. Picture (Owner of Gym and Me)

A short-recorded video interacting with the owner and some photos: [Drive Link](#)

3. Metadata and Descriptive Statistics

a) Metadata

I have collected the data from 01 January 2024 to 31 August 2024 in four sheets. Sheets contains:

- **Demographics Data:** Member ID, Membership Type, Age, Gender, Address, Distance from gym in km, Fees in Rupees per month
- **Member Behavior Data:** Member ID, Check-In Time, Check-Out Time, Duration (hrs.)
- **Member Distribution:** Member's Gender Distribution over months, Membership subscription type distribution
- **Overcrowding Data:** Time Slot, Area/Zone, Equipment in Use, Number of Members, Comments

LINK for Data Set: [DATA SET](#)

Member ID	Membership Type	Age	Gender	Address	Distance from Gym (km)	Fees (Rupees) per month
1	1 Month	16	Male	Pochampur Village	1	1200
2	2 Month	16	Male	Dwarka Sector 22	3	1200
3	3 Month	16	Male	Pochampur Village	1	1200
4	4 Annual	17	Male	Dwarka Sector 23	2	1100
5	5 Quarterly	17	Male	Dwarka Sector 21	3.5	1150
6	6 Annual	18	Male	Dwarka Sector 22	3	1100
7	7 Month	18	Male	Dwarka Sector 21	3	1200
8	8 Annual	18	Male	Pochampur Village	1	1100
9	9 Annual	18	Male	Dwarka Sector 21	3.5	1150
10	10 Quarterly	18	Male	Dwarka Sector 22	3	1150
11	11 Quarterly	19	Male	Pochampur Village	1	1150
12	12 Quarterly	19	Male	Pochampur Village	1	1150
13	13 Quarterly	19	Male	Pochampur Village	1	1150
14	14 Quarterly	19	Male	Pochampur Village	1	1150
15	15 Quarterly	19	Male	Pochampur Village	1	1200
16	16 Quarterly	19	Male	Dwarka Sector 23	2	1200
17	17 Annual	19	Male	Dwarka Sector 23	2	1100
18	18 Annual	19	Male	Dwarka Sector 23	3	1100
19	19 Quarterly	19	Male	Dwarka Sector 21	3.5	1200
20	20 Monthly	20	Male	Dwarka Sector 21	3.5	1200
21	21 Monthly	20	Male	Pochampur Village	1	1200
22	22 Annual	20	Male	Pochampur Village	1	1100
23	23 Quarterly	20	Male	Pochampur Village	1	1150
24	24 Annual	20	Male	Pochampur Village	1	1100
25	25 Monthly	20	Male	Pochampur Village	1	1200
26	26 Annual	20	Male	Dwarka Sector 21	3.5	1100
27	27 Annual	21	Male	Dwarka Sector 21	3.5	1100
28	28 Quarterly	21	Male	Dwarka Sector 23	2	1150
29	29 Quarterly	21	Male	Dwarka Sector 23	2	1150
30	30 Quarterly	21	Male	Dwarka Sector 23	2	1150
31	31 Quarterly	22	Male	Pochampur Village	1	1150
32	32 Quarterly	22	Male	Pochampur Village	1	1150
33	33 Monthly	22	Male	Pochampur Village	1	1200
34	34 Monthly	22	Male	Dwarka Sector 22	3	1200
35	35 Annual	22	Male	Dwarka Sector 22	3	1100
36	36 Annual	22	Male	Pochampur Village	1	1100
37	37 Monthly	22	Male	Pochampur Village	1	1200
38	38 Monthly	23	Male	Pochampur Village	1	1200
39	39 Monthly	23	Male	Pochampur Village	1	1200
40	40 Annual	23	Male	Pochampur Village	1	1100
41	41 Quarterly	23	Male	Pochampur Village	1	1150
42	42 Annual	23	Male	Dwarka Sector 22	3	1200
43	43 Monthly	23	Male	Dwarka Sector 22	3	1200
44	44 Annual	23	Male	Dwarka Sector 22	3	1100

Fig 3. Collected Data from the business

b) Descriptive Statistics

Total fee amount collected in 8 months: ₹11,61,600

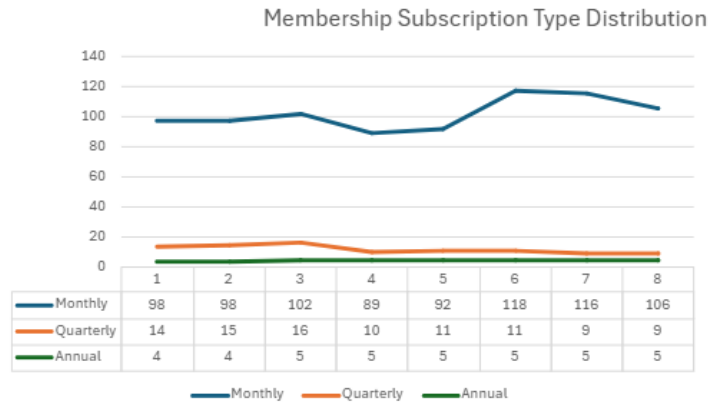


Fig 4. Membership Subscription Type Distribution

The above line chart will help us to find out the membership subscription type preferred by the members.

Most preferred membership subscription type: Monthly because many people try, and then within a month they get bored.

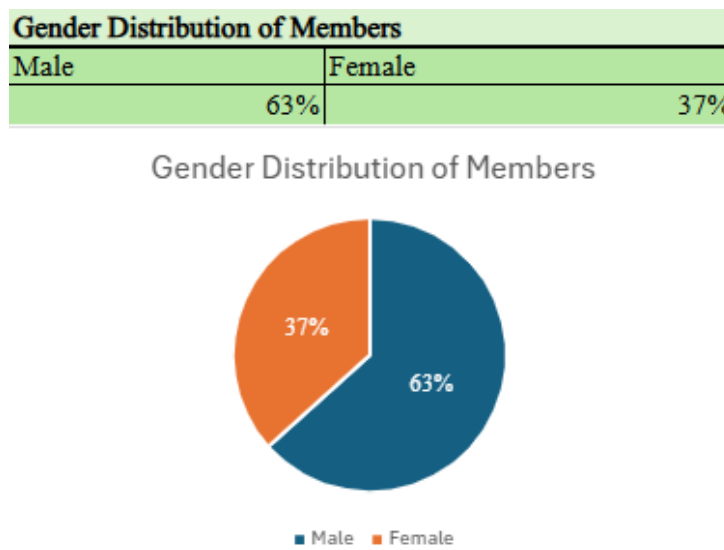


Fig 5. Pie Chart representing Gender Ratio of Members for the Month (August)

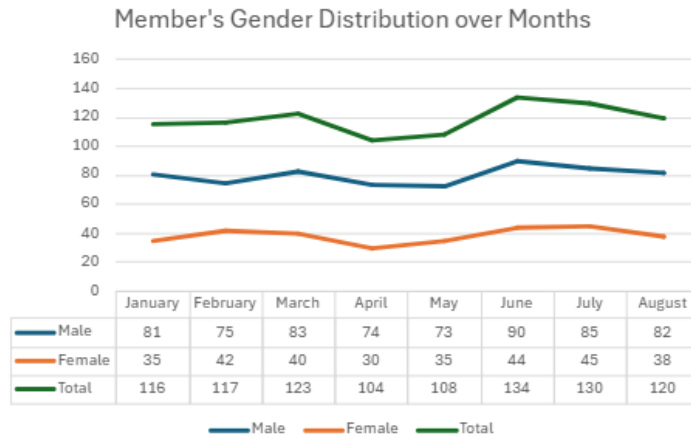


Fig 6. Member's Gender Distribution over a period of 8 months

The above line chart will help us to find out the peak months of the business.

Peak months of business: March, June, July because of schools and colleges vacations.

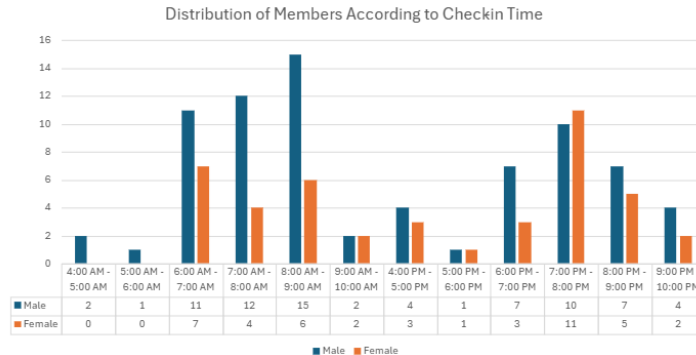


Fig 7. Distribution of Members According to Check-in Time

The above chart will help us to find out the peak hours and non-peak hours of the day to solve the overcrowding problem at certain period of times.

Peak hours of the day: 6 AM – 9 AM and 6 PM – 9 PM as many people want to exercise before going to the office or after returning from work. The most crowded time is typically between 7 pm and 8 pm in the evening, as busy women during the day and late-night diners tend to come during this time, along with some couples or siblings who also want to exercise.

4. Detailed Explanation of Analysis Process / Method

Data for 8 months was collected from the business in Microsoft Excel which was then cleaned and managed for further use.

Data such as Demographic data, Member Behavior data, Member Distribution over 8 months and Overcrowding data was collected. Unnecessary data has been removed, ensuring that the relevant data remains.

Analysis Process:

Line chart - Membership Subscription Type Distribution

- Plot the Membership Subscription Type against Period of time to visualize the preferred membership subscription type by members.

Pie Chart - Gender Ratio of Members for the Month (August)

- Create a pie chart showing the distribution of Gender Ratio of Members for the Month (August)

Line Chart - Member's Gender Distribution over a period of 8 months

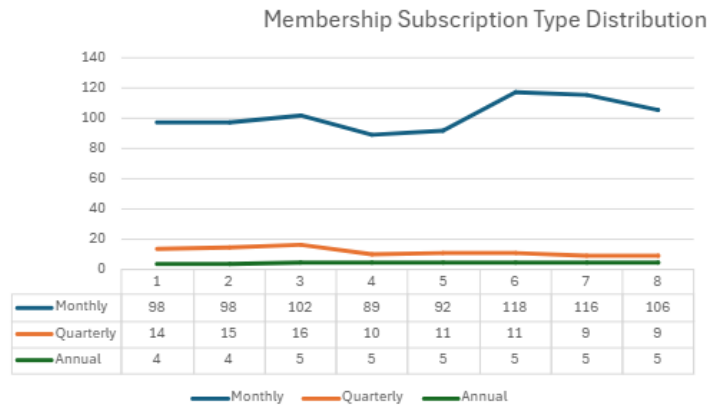
- Plot the Member's Gender distribution and number of total members against Months to analyze any noticeable patterns, such as seasonal fluctuations.
- To find out the peak months of the business.

Column Chart - Distribution of Members According to Check-in Time

- Plot the number of members against check-in time.
- To find peak hours of the day.

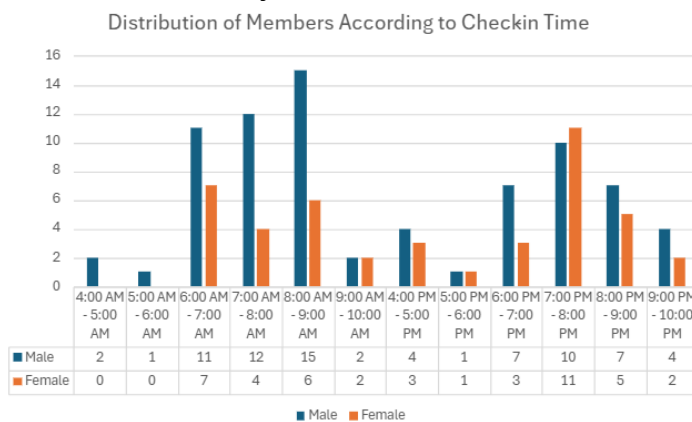
5. Results and Findings

- Membership Retention Insights:



Members with monthly membership subscription type shows a higher dropout rate than members with quarterly and annual membership subscription type, suggesting a need for improved engagement and retention strategies for short-term memberships.

- Peak Hours of the Day:



During the gym working hours, the most crowded time is 6 AM – 9 AM in the morning and in the evening, it is 6 PM – 9 PM. So, the gym owner should provide discount on membership subscription to members who come during off-peak hours, can get special attention from trainers who come in off peak to reduce overcrowding at specific times, which will also assist in membership retention and acquisition.

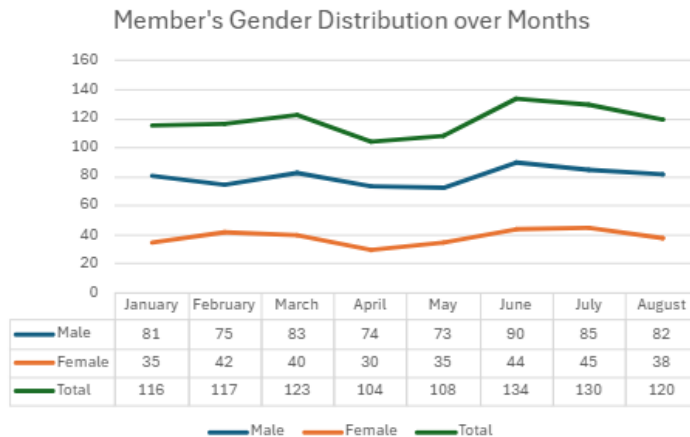
- Member Satisfaction Levels:



Members have expressed dissatisfaction with the availability of equipment in peak hours, necessitating improvements in operations.

Members Acquisition is also affected by Members Satisfaction Level and these ratings are according to the overcrowding data against time slot.

- Peak Months of business:



Membership increases in March, June and July because schools and colleges vacations as school exams end in March and students showcase their interests in the gym, and college exams ends in June and July.